

Joshua W. Z. Hew

📍 Berkeley ✉ joshua.hew@berkeley.edu 🔗 <https://j2hwz.github.io/> in J2hwz 📷 J2hwz

Education

University of California, Berkeley

Aug 2026 –

PhD in Psychology

- Advisor: Marina Dubova

University College London

Sep 2025 – Aug 2026

MSc Cognitive and Decision Sciences (expecting First Class)

- Dissertation: “Evaluating the Instrumental Goals of Large Language Models” (submitted). Advisor: Mario Giulianelli

University of Edinburgh

Sep 2021 – May 2025

MA (Hons) Philosophy and Psychology (First Class Honours)

- Dissertation: “Learned helplessness in dynamic causal environments: The impact of controllability and reward prevalence” (85%). Advisor: Neil Bramley

Bridging Programme (UCL Gatsby Computational Neuroscience Unit)

June 2026 – Aug 2026

Intensive mathematics summer school for research in theoretical neuroscience and machine learning.

- **Topics:** linear algebra, multivariate calculus, statistics, dynamical systems, Fourier analysis, & convolution.

Publications

Hew, J. W. Z., & Bramley, N. R. (2026). Causal inference and learned helplessness. In *Proceedings of the 48th Annual Meeting of the Cognitive Science Society*.

Hew, J. W. Z., Tarighat, A., Corley, M., & Horne, Z. (2024). Examining structural and semantic predictors of announced sarcasm on r/AskReddit. In *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*.

In preparation:

Hew, J. W. Z., Rottman, J., & Horne, Z. The language of doubt impacts consensus judgements.

Presentations & Talks

Causal inference and learned helplessness

- Talk presented at Bramley Computational Cognitive Science Lab (March 2025).
- Talk presented at British Psychological Society Undergraduate Conference, University of Abertay (March 2025).
- Talk presented at Edinburgh Cognitive Science Conference (June 2026).
- Poster presented at Cognitive Science Conference, Rio de Janeiro (July 2026).

The language of doubt impacts consensus judgements

- Talk presented at Cognition, Computation and Development Lab, University of Edinburgh (October 2024).

Examining structural and semantic predictors of announced sarcasm on r/AskReddit

- Poster presented at Social Data Science Poster Session, University of Edinburgh (June 2024).
- Poster presented at Cognitive Science Conference, Rotterdam (July 2024).
- Talk presented at Brain, Cognition and Development Seminar, University of Edinburgh (Nov 2024).

Awards

Berkeley Fellowship for Graduate Study (\$41000)	2026
Carnegie Trust Summer Vacation Scholarship (£3800)	2024
University of Edinburgh Our Minds Scholarship (£1600)	2024

Research Experience

Cognition, Computation and Development Lab (PI: Zachary Horne) **University of Edinburgh**
Research Assistant *Jan 2024 - Apr 2025*

- Administered 2 online experiments exploring how prior beliefs influence consensus judgements, using real-world agreement statistics from Pew Research. Discovered that information framed to imply doubt reduces the influence of prior beliefs, guiding subsequent studies on behavioural responses to aggregated information.
- Conducted an experiment to probe participant intuitions about the distributions of events and beliefs. Used this to investigate whether people systematically underestimate the probability of these distribution yielding overlapping evidence by chance.
- Applied Huggingface models and cosine similarity methods to quantify the uniqueness of comments on r/Showerthoughts. Used this as a naturalistic measure of how likely random search within internet communities generates overlapping (2+) evidence that could spur belief change.

Independent Psycholinguistics Project (Supervisor: Martin Corley) **University of Edinburgh**
Research Assistant *Sep 2024 - Apr 2025*

- Created and pretested a set of 32 stimuli, analysed results using ordinal regression models.
- Designed and deployed an online self-paced reading study in PCIBex to investigate sarcasm processing when reading direct or indirect speech.

User-centric Privacy and Security Research Team (PI: Jingjie Li) **University of Edinburgh**
Research Intern *Mar - Aug 2024*

- Collaborated with team members to develop a theory-informed taxonomy of unsafe content.
- Used this taxonomy to code 80 sections of script from a role-playing game.

Holly-Martin Psycholinguistics Lab (Supervisor: Greta Gandolfi) **University of Edinburgh**
Research Assistant *Autumn 2023*

- Annotated the linguistic features for transcriptions of conversations between humans and social robots.
- Coded an R script to check for annotator spelling mistakes and calculate inter-annotator agreement.

Teaching Experience

Big Data and Psychological Science **University of Edinburgh**
Teaching Assistant *Autumn 2024*

- Assisted students with coding exercises in R, and provided advice on final projects.
- Developed a reference guidebook using R Markdown containing coding tutorials and course tips.

Other Relevant Experience

Undergraduate Psychology Association (UPA) **United Kingdom**
Journal Web Manager *Oct 2025 - Present*

- Supported the establishment of the association's journal, with a focus on web design.

Edinburgh Nightline **Edinburgh, UK**
Coordinator, Research and Policy Coordinator *Apr 2023 - Apr 2025*

- Managed committee and various projects, communicated with stakeholder institutions and trustees.
- Reviewed industry best practices to make improvements and facilitated research-informed focus groups.
- Visualised and interpreted call data and content to improve service quality and demonstrate impact.
- Former shift-taking listening volunteer, applying active listening and non-directionality techniques.

Psychology Magazine **University of Edinburgh**
Head writer, Writer *Spring 2023, 2024*

- Wrote 2 research articles aimed at the general public: "Can Money Buy Happiness?" (2023), "The Good, the Bad, and the Obsession: Understanding Stan Culture in the Digital Era" (2024).
- Offered peer support and organised writing workshops.

Additional Information

Reviewing: Cognition (1), Cognitive Science Conference Proceedings (1).

Programming: R, Python, HTML, CSS, Javascript, jsPsych, MATLAB

Misc: Qualtrics, LaTeX, PCIBex, Adobe Premiere Pro, FlipaClip (animation).

Languages: English (native), Chinese (fluent), Malay (proficient)

Personal Projects: YouTube channel with 6 videos with 3000+ views, aim to create educational and scientific animated videos in the future. *Lepak Time* podcast ran with friends during COVID consisting 21 episodes.

Other Interests: Piano (diploma), Violin (grade 8), Bildungsroman literature, cycling, and bouldering.